

VOLUNTEERHUB

Volunteer & Social Service Activity Management Portal

PROJECT DOCUMENTATION

CodeFusion 2026

Project Information	Details
Project Name	VolunteerHub
Project Type	Web-based Volunteer & Social Service Activity Management Portal
Primary Roles	Admin, Organizer / Staff, Volunteer / User
Frontend	React (Next.js App Router), Tailwind CSS, shadcn/ui
Backend	Next.js API Routes / Node.js
Database & Authentication	Supabase (PostgreSQL, Row-Level Security, Realtime, Storage)
Deployment	Vercel
Version	1.0
Prepared By	Team - FusiO(n) Squad
Institution	Bannari Amman Institute of Technology

Document Control

Field	Entry
Document Title	VolunteerHub Project Documentation
Project	Volunteer & Social Service Activity Management Portal
Date	1 August 2026

Repository & Project Links:

Vercel Link: <https://impact-pulse-final.vercel.app/>

GitHub Link: <https://github.com/bharani-thangavel/ImpactPulse>

Project UI – Front Page :

ImpactPulse Supabase Live
Volunteer Impact Management Platform

Light Sign In / Register: Admin Organizer Volunteer

Empowering Trusted Volunteering Ecosystems

Connecting Purpose to Measurable Impact

ImpactPulse is the enterprise SaaS platform for verified event coordination, automated attendance tracking, gamified rewards, and AI-driven impact analytics.

[Get Started as Volunteer →](#) [Register Organization](#) [System Admin Portal](#)

1,450+ VOLUNTEERS	12,800+ HOURS TRACKED	320+ EVENTS CONDUCTED	98.4% VERIFICATION RATE
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1. Project Overview

VolunteerHub is a web-based Volunteer and Social Service Activity Management Portal designed to centralize and digitize the lifecycle of volunteering activities. The system supports administrators, organizers/staff, and volunteers/users through role-specific workflows for event creation, approval, registration, attendance, leadership selection, notifications, feedback, points, badges, and reporting.

The platform is intended to reduce manual paperwork and communication delays while improving transparency, accountability, collaboration, and the overall volunteering experience.

1.1 Project Summary

Aspect	Description
Problem Domain	Volunteer and social service activity management
Core Approach	Centralized role-based web portal
Main Workflow	Create → Approve → Publish → Register → Participate → Track → Recognize
Primary Roles	Admin, Organizer / Staff, Volunteer / User
Key Differentiators	Leadership workflow, skill tagging, slot-based signups, waitlists, recurring activities, digital consent, points and badges

1.2 Project Abstract

VolunteerHub provides a centralized platform for managing social service and volunteering activities. It replaces fragmented manual processes with role-based digital workflows. Administrators can oversee organizers and events, organizers can create and manage activities and select leaders, and volunteers can discover activities, register, monitor participation, and receive recognition through points and badges. The system also provides notification, feedback, and reporting capabilities to improve institutional visibility.

2. Problem Statement

The existing volunteering process can involve manual records, fragmented communication, and limited visibility into event availability and participation. VolunteerHub addresses these limitations by digitizing the complete activity-management lifecycle.

2.1 Major Problems Identified

- Manual, paper-based tracking of volunteer activities and attendance.
- No centralized system for approving events or organizers.
- Lack of real-time visibility into available activities and seat capacity.
- No structured way to recognize or reward volunteer contribution.
- Absence of analytics/reporting for institutional accountability.

2.2 Proposed Solution

VolunteerHub introduces a centralized portal that connects administrators, organizers, and volunteers. It streamlines event creation and approval, registration, attendance tracking, leadership selection, notifications, feedback, recognition, and reporting.

3. Objectives

- Simplify volunteer registration and participation.
- Enable organizers to efficiently create and manage events.
- Provide administrators with centralized approval and monitoring capabilities.
- Maintain accurate attendance and participation records.
- Recognize volunteers through points, badges, and certificates.
- Improve communication through real-time notifications.
- Generate reports and analytics for better decision-making.

4. Target Users & Roles

Role	Access Level	Primary Capabilities
Admin	System-wide	Approve/decline events and organizers; search organizer and volunteer lists; audit and oversee platform activity.
Organizer / Staff	Event-scoped	Create events; manage ongoing events and history; select team leaders; view member and leader details.
Volunteer / User	Self-scoped	Browse and register for activities; track attendance and history; view profile, badges, points; receive notifications.

5. Scope of the System

The documented scope includes the following major areas:

- Role-based registration, authentication, and access control.
- Organizer and event approval workflows.
- Event creation, publishing, ongoing-event management, completion, cancellation, and archival lifecycle.
- Slot/shift-based volunteer signups and seat-capacity tracking.
- Volunteer registration and attendance tracking.
- Leadership application and selection.
- Skill tagging for volunteers and activity requirements.
- Waitlists and auto-promotion when capacity becomes available.
- Recurring activities.
- Digital consent/waiver forms.
- Notifications, feedback, points, badges, certificates, and participation history.
- Administrative search, monitoring, reports, and analytics.

6. Functional Requirements

ID	Requirement	Description
FR-01	User registration and role-based authentication	Users can register and access role-specific functionality.
FR-02	Event creation and management	Organizers can create and manage volunteering events.
FR-03	Event approval	Administrators can approve or decline events.
FR-04	Event registration	Volunteers can browse and register for available events.
FR-05	Attendance tracking	Organizers/assigned leaders can mark attendance.
FR-06	Leadership application and approval	Volunteers can apply for leadership and selected leaders can be assigned.
FR-07	Notifications	Users can receive activity-related notifications.
FR-08	Feedback	Participants can submit comments and ratings.
FR-09	Rewards	Volunteer points and badges can be maintained.
FR-10	Administration	Admins can manage users, events, and reports.

7. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Fast response and support for concurrent users.
NFR-02	Security	Secure authentication and role-based access.
NFR-03	Reliability	Accurate and consistent data handling.
NFR-04	Usability	Responsive and user-friendly interface.
NFR-05	Scalability	Support future growth in users and events.
NFR-06	Maintainability	Modular and well-structured architecture.

8. Technology Stack

Layer	Technology	Purpose
Frontend	React + Next.js App Router	Build responsive role-specific interfaces.
UI	Tailwind CSS + shadcn/ui	Styling and reusable UI components.

Backend	Next.js API Routes / Node.js	Business logic and API endpoints.
Database	Supabase PostgreSQL	Store users, events, registrations, notifications, feedback, and rewards.
Authentication	Supabase Auth / role-based access	Secure sign-in and authorization.
Security	Row-Level Security	Restrict data access according to roles and ownership.
Realtime	Supabase Realtime	Real-time updates/notifications.
Storage	Supabase Storage	File/storage support where required.
Email	Resend / Nodemailer	Email notifications.
Deployment	Vercel	Deploy the web application.

9. System Architecture & Design

The system follows a web application architecture in which the Next.js frontend communicates with backend API routes/Node.js services and Supabase provides PostgreSQL database, authentication, Row-Level Security, Realtime, and Storage capabilities.

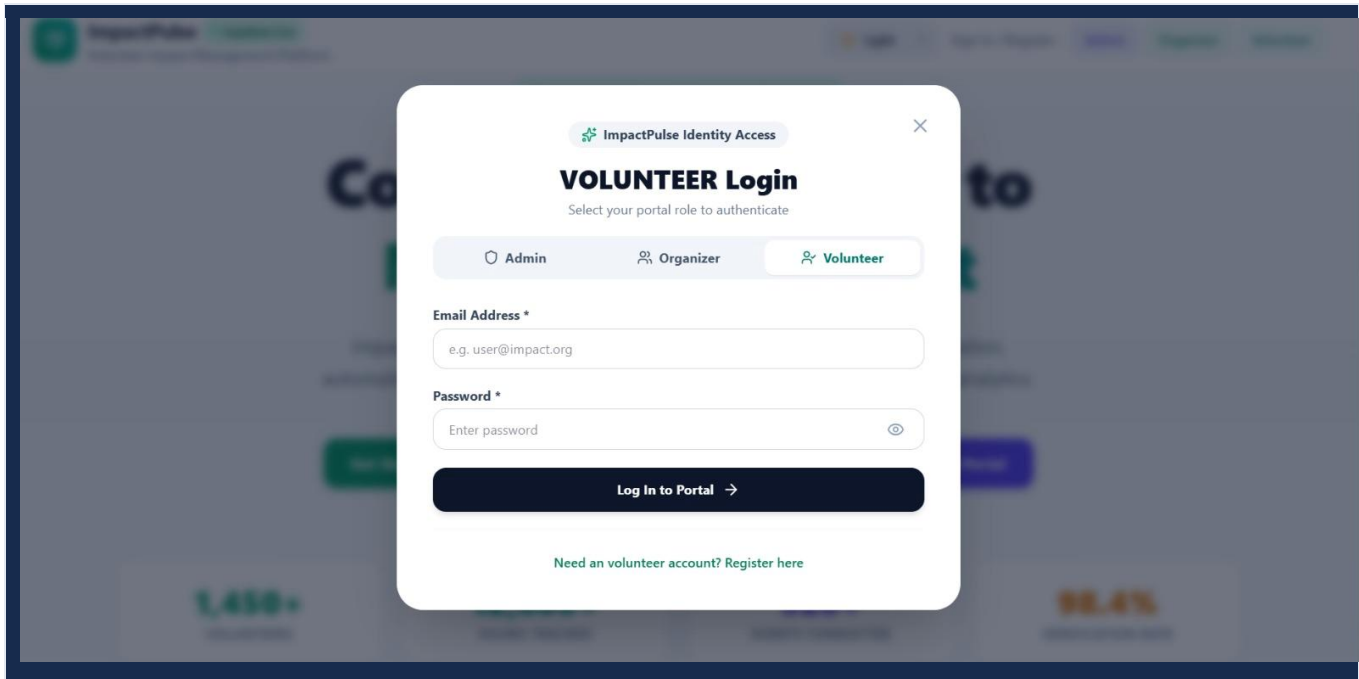
9.1 Role-Based Architecture

Role	Main Flow
Admin	Login → Admin Dashboard → Approve organizers/events → Search users → Monitor activities → Reports/Oversight
Organizer	Signup → Admin Approval → Organizer Dashboard → Create Event → Manage Event → Select Leader → Track Attendance
Volunteer	Signup → Volunteer Dashboard → Browse Activities → Register/Waitlist → Attend → Feedback → Points/Badges/History

10. User Workflows

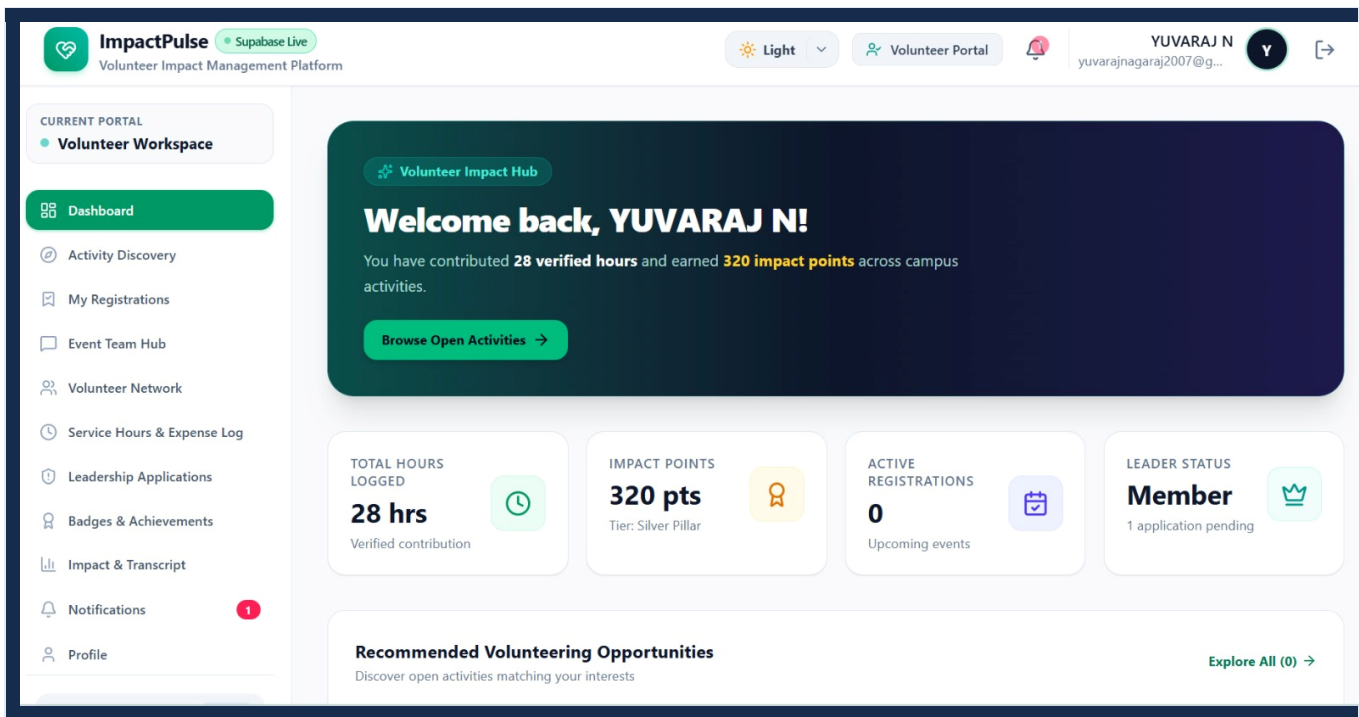
10.1 Volunteer / User Signup Flow

1. User provides credentials such as email and password.
2. User enters interest/profile information such as name, age, and other requirements.
3. User indicates interest in leadership or direct volunteering.
4. Leadership interest is routed through an approval workflow.
5. Direct volunteering routes the user into the Volunteer role.



10.2 Organizer Signup Flow

6. Organizer provides credentials.
7. Organizer submits organizational information such as registration number, department, and year.
8. Application is routed to Admin for approval.
9. After approval, the organizer receives access to the Organizer Dashboard.



10.3 Event Participation Flow

10. Organizer creates an event.
11. Admin approves the event where required.

12. Event becomes available to volunteers.
13. Volunteer selects an available activity/slot and registers.
14. If capacity is reached, the volunteer may be placed on a waitlist.
15. Attendance is recorded during the event.
16. Participation can result in feedback, points, badges, and certificates.

11. Module & Feature Documentation

11.1 Authentication & Role Management

- Registration and login.
- Role-based access for Admin, Organizer/Staff, and Volunteer/User.
- Approval status for organizer and applicable user workflows.
- Secure authentication and authorization.

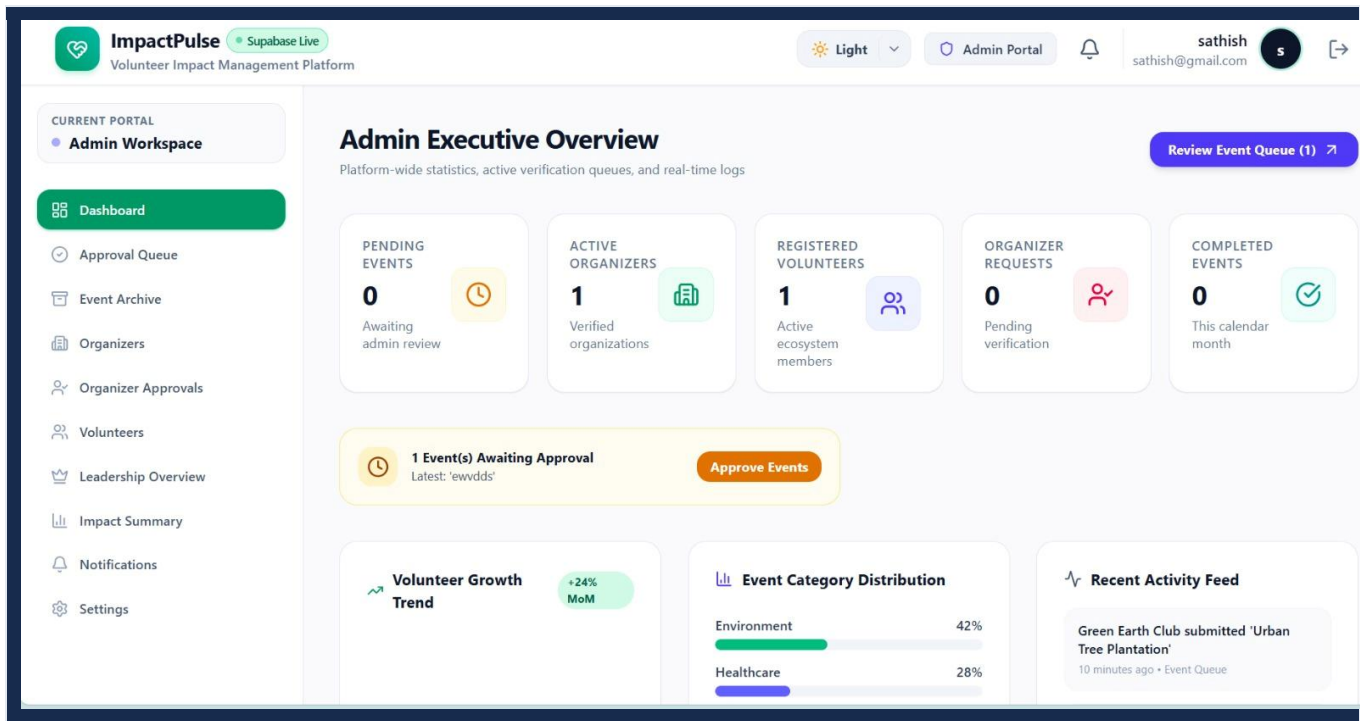
11.2 Organizer Dashboard

Tab	Features
Create Event	Title, description, members required, timing/date, duration, venue, contact details, points rewarded, available seats.
My Events	Ongoing events with member/leader details; history of past events and event status.
Leader Selection	View leadership applicants for organizer events and assign selected volunteers as leaders.

The screenshot displays the ImpactPulse Organizer Dashboard for Organization1. The interface includes a sidebar with navigation options like Dashboard, Create Event, My Events, History, Attendance, Leader Selection, Burnout Early-Warning, Impact Summary, Notifications, and Profile. The main content area features a top navigation bar with the ImpactPulse logo, a 'Scout Quick Notification' button, and a 'Create New Event' button. Below this, there are several key metrics: Active Events (1), Pending Approvals (0), Leader Applicants (0), Seat Fill Rate (0%), and Completed Events (0). A 'Quick Actions' section provides shortcuts for Quick Broadcast, Create Event, Attendance, Leader Selection, Burnout Alerts, and Impact Summary. The 'Upcoming Schedule & Active Events' section shows an event titled 'CODE CIRCLE' with a status of 'ongoing' and '0/20 Joined'. The 'Pending Leader Applicants' section indicates 'No pending leadership applications'.

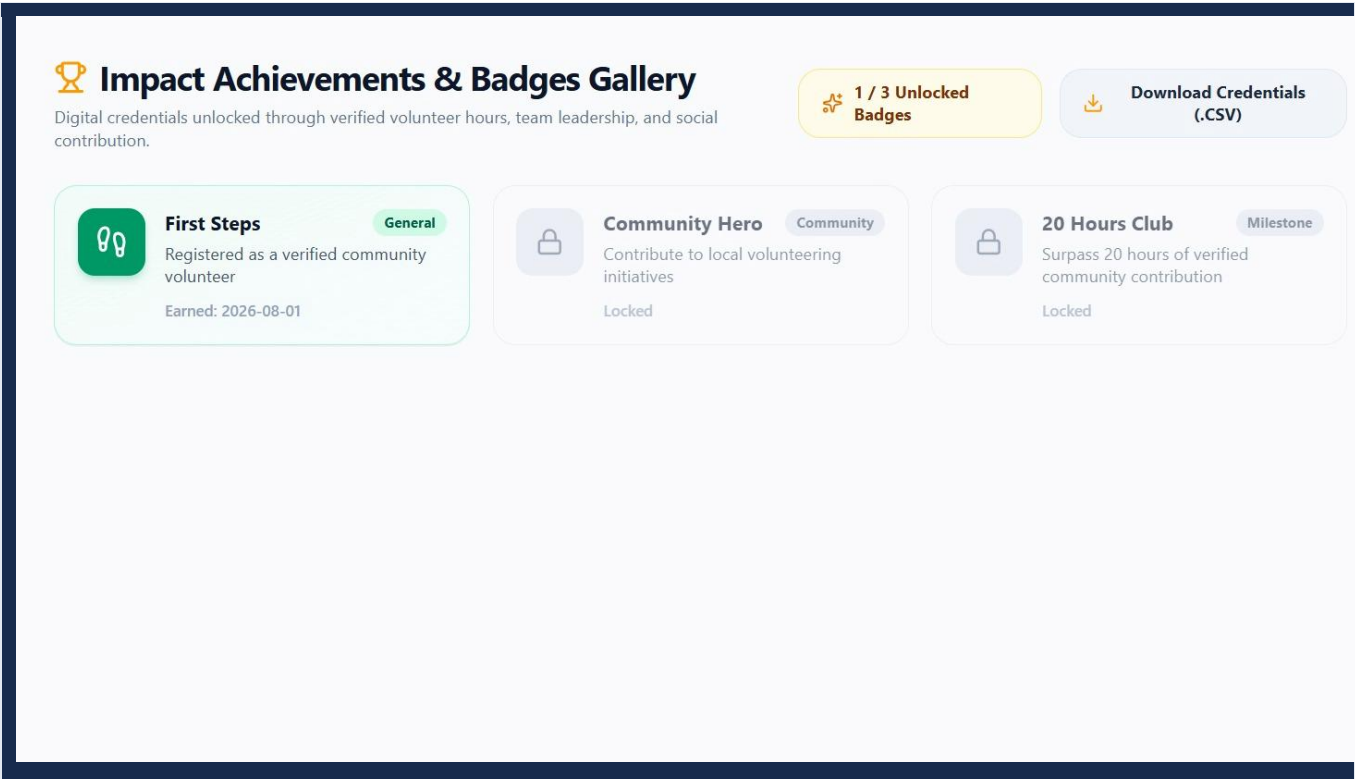
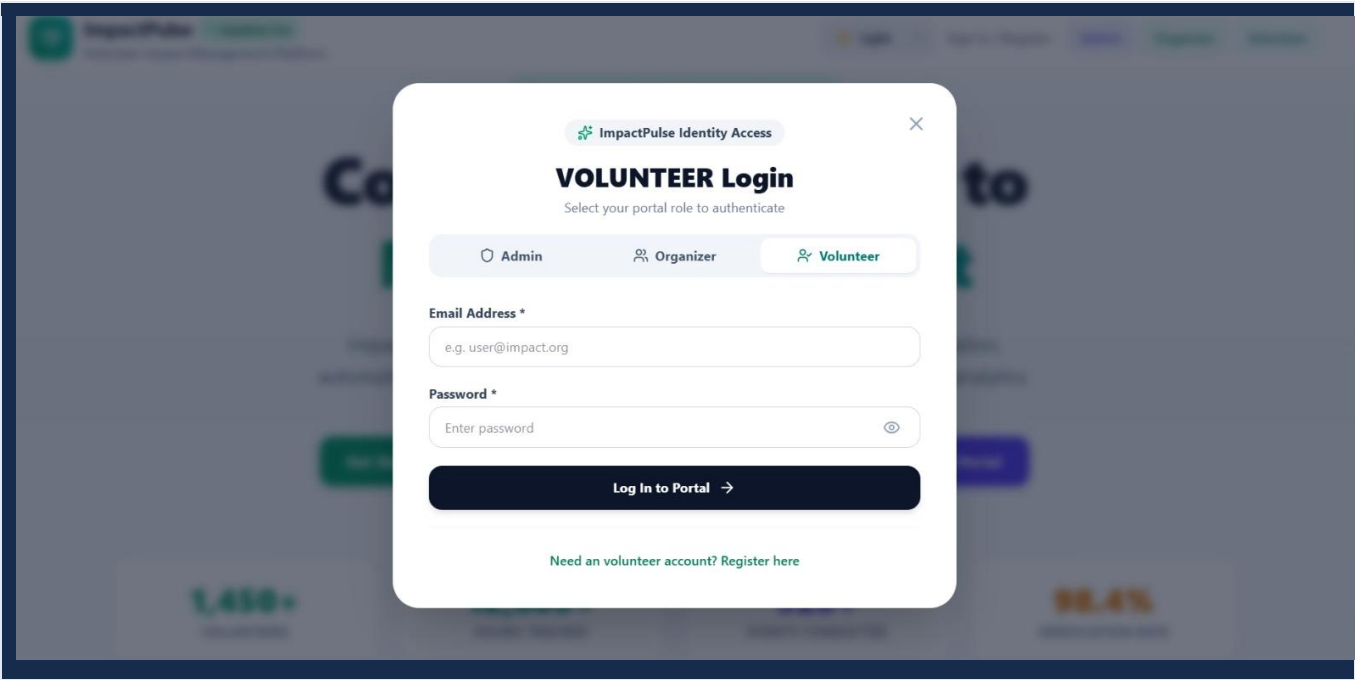
11.3 Admin Dashboard

Tab	Features
Event List	View current events, approve/decline pending events, and track event status.
Organizer's Tab	Searchable list of organizers and their credentials.
Volunteer's Tab	Searchable list of volunteers and their credentials.



11.4 Volunteer / User Dashboard

Tab	Features
Homepage	Floating task shortcuts, notification feed, points summary widget.
Booking	Available activities, registration, history, and attendance records.
Profile	Personal details, badges, total points, member-since date, and total volunteering hours.



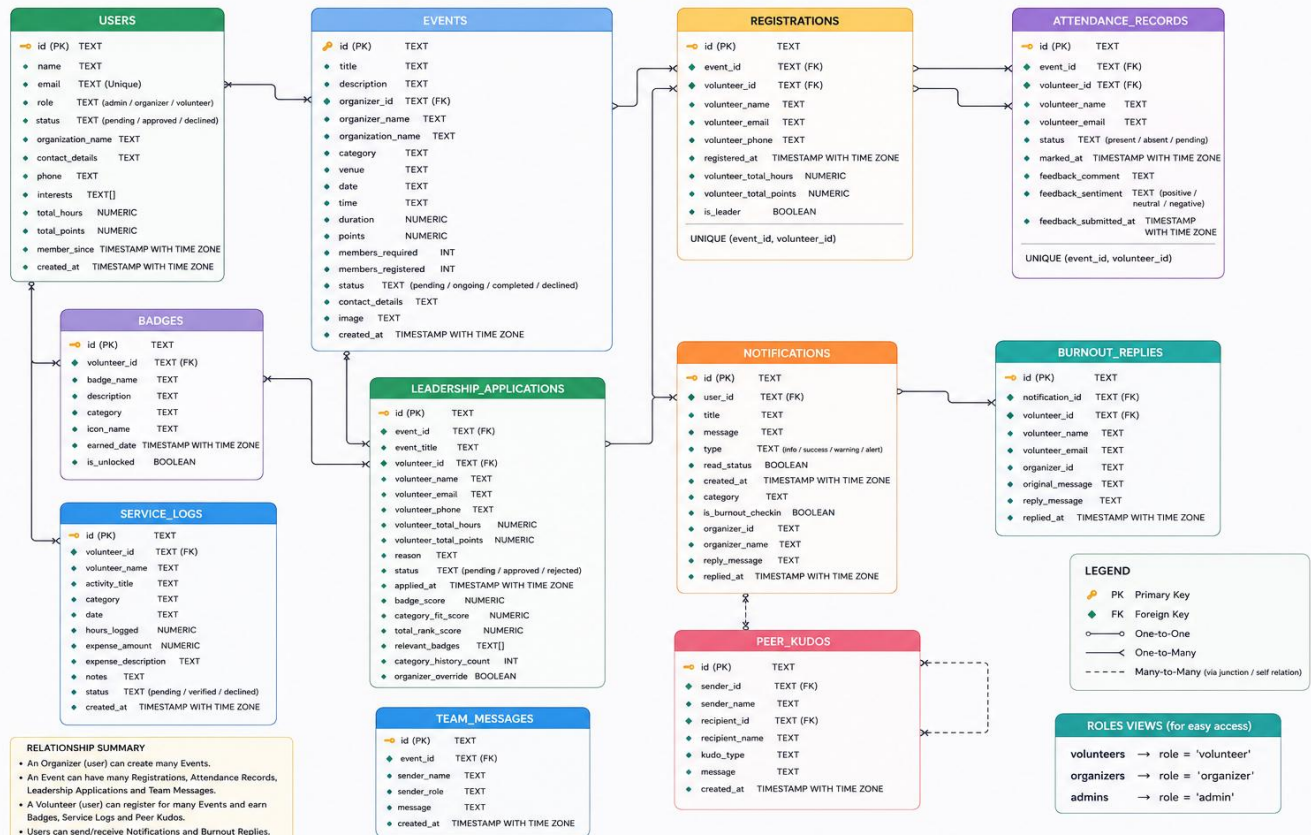
12. Database Design

The core entities specified for the platform are:

Entity	Key Fields / Purpose
Users	id, name, email, password_hash, role, age, reg_no, department, year, status

Events	id, organizer_id, title, description, members_required, timing, date, duration, venue, contact_details, points_rewarded, status, available_seats
Registrations	id, event_id, user_id, registration_date, attendance_status, role_in_event
LeadershipApplications	id, user_id, event_id, status, badge_awarded
Notifications	id, user_id, message, type, is_read, created_at
Feedback	id, event_id, user_id, comments, rating
Badges/Points	id, user_id, total_points, member_since, badges_earned

ImpactPulse2.0 – Database ER Diagram



12.1 Database Screenshot / Supabase Evidence



13. API Documentation

Module	Method	Endpoint	Purpose
Authentication	POST	/api/auth/signup	Register a new user with role-based access.
Authentication	POST	/api/auth/login	Authenticate user and generate access token.
Events	GET	/api/events	Retrieve events with optional filters.
Events	POST	/api/events	Create a new event (Organizer).
Events	PATCH	/api/events/:id/status	Approve, decline, or cancel an event (Admin).
Registration	POST	/api/events/:id/register	Register a volunteer for an event.
Attendance	PATCH	/api/events/:id/attendance	Mark volunteer attendance.
Leadership	POST	/api/leadership/apply	Submit a leadership application.
Leadership	PATCH	/api/leadership/:id/approve	Approve/reject leadership application.
Notifications	GET	/api/notifications	Retrieve notifications for authenticated user.
Admin	GET	/api/admin/organizers	Retrieve/search organizer records.
Admin	GET	/api/admin/volunteers	Retrieve/search volunteer records.

13.1 API Testing Evidence

[API TESTING SCREENSHOT]

Insert Postman/Thunder Client/API client screenshots showing successful requests and responses.

Replace this placeholder with the actual project image/screenshot before final submission.

API Collection / Documentation Link: [\[Paste link here\]](#)

14. UI / UX Documentation

This section is reserved for documenting the final user interface. Add screenshots of each completed screen and briefly describe what the user can do on that screen.

Screen	Screenshot	UI Description
Login Page	[Insert screenshot]	[Describe login/authentication behavior]
Volunteer Signup	[Insert screenshot]	[Describe onboarding fields and choices]
Organizer Signup	[Insert screenshot]	[Describe organization details and approval flow]
Admin Dashboard	[Insert screenshot]	[Describe event/user oversight]
Organizer Dashboard	[Insert screenshot]	[Describe event management and leader selection]
Volunteer Homepage	[Insert screenshot]	[Describe notifications, shortcuts, points]
Booking Page	[Insert screenshot]	[Describe event discovery and registration]
Profile Page	[Insert screenshot]	[Describe profile, badges, points, hours]

15. Testing & Validation

Record the major test cases used to verify the implemented system.

ID	Test	Input / Condition	Expected Result	Status
TC-01	Valid login	Correct credentials	User reaches correct role dashboard	Pass / Fail
TC-02	Invalid login	Incorrect credentials	Error message shown; access denied	Pass / Fail
TC-03	Organizer event creation	Valid event details	Event is created and enters expected workflow	Pass / Fail
TC-04	Admin event approval	Pending event	Admin can approve/decline event	Pass / Fail
TC-05	Volunteer registration	Open event with seats	Registration succeeds and seat count updates	Pass / Fail

TC-06	Attendance tracking	Registered volunteer	Attendance status is recorded	Pass / Fail
TC-07	Leadership workflow	Leadership application	Application can be reviewed/approved	Pass / Fail
TC-08	Notifications	User activity	Relevant notification is delivered/displayed	Pass / Fail
TC-09	Profile rewards	Completed participation	Points/badges/history reflect expected result	Pass / Fail
TC-10	Role access	Different user roles	Unauthorized role cannot access restricted features	Pass / Fail

16. Innovation / Novelty

- Role-based workflow mirrors institutional processes and supports transparency and accountability.
- Leadership application and approval creates a growth path for volunteers beyond simple event registration.
- Points, badges, and volunteering-hour tracking provide a gamified recognition layer.
- Shift/slot-based signups allow volunteers to select specific time slots.
- Skill tagging can connect volunteer capabilities with activity requirements.
- Waitlists can automatically promote volunteers when capacity becomes available.
- Recurring activities can be managed without recreating the same event each time.
- Digital consent/waiver forms support institutional and compliance-oriented workflows.
- The planned AI/analytics layer can support intelligent matching and reporting as the project evolves.

17.1 Innovation Evidence

[INNOVATION / ANALYTICS SCREENSHOT]

Insert the implemented innovative feature, analytics view, or AI-assisted feature here.

Replace this placeholder with the actual project image/screenshot before final submission.

18. Future Enhancements

- Expand intelligent volunteer-to-activity matching based on skills and interests.
- Enhance analytics dashboards for institutional reporting.
- Add richer certificate generation and verification.
- Expand recurring event automation and scheduling.
- Integrate additional institutional systems where appropriate.
- Improve notification channels and configurable reminders.

19. Team Details

Member	Reg. No.
Bharani T	7376252AD131

Yaswanth Kalis P	7376252AD377
Sri Satish B	7376252AD342
Yuvaraj N	7376252AD380

20. Conclusion

VolunteerHub digitizes the volunteer management lifecycle from event creation and approval through discovery, registration, attendance, leadership selection, feedback, and recognition. The role-based architecture provides a clear separation of responsibilities between administrators, organizers, and volunteers while supporting transparency and efficient collaboration. The project documentation can be updated with the final implementation screenshots, GitHub links, API evidence, database evidence, and deployment details before submission.

Final GitHub Link: <https://github.com/bharani-thangavel/ImpactPulse>